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Hierarchical Algorithms
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Exercise sheet 07, Solution hints
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Homework:

Exercise 1: Let $p \in \mathbb{N}$. Let the $n = 2^p$ points $x_i = i \cdot h$ for $i \in \{1, \dots, n\}$ and $h = \frac{1}{n+1}$ be given in the interval $[0, 1]$. For $\ell \in \{0, \dots, p\}$ and $j \in \{1, \dots, 2^{p-\ell}\}$, let X_j^ℓ be the set of 2^ℓ consecutive points starting with $x_{(j-1)\cdot 2^\ell+1}$, i.e.

$$\begin{aligned}
X_1^0 &= \{x_1\}, & \cdots, \quad X_n^0 &= \{x_n\},\\
X_1^1 &= \{x_1, x_2\}, & \cdots, \quad X_{\frac{n}{2}}^1 &= \{x_{n-1}, x_n\},\\
&\vdots\\
X_1^{p-1} &= \{x_1, \dots, x_{n/2}\}, & X_2^{p-1} &= \{x_{n/2+1}, \dots, x_n\},\\
X_1^p &= \{x_1, \dots, x_n\}.
\end{aligned}$$

We set

$$\begin{aligned}
x_{j,\ell}^s &:= x_{(j-1)\cdot 2^\ell+1} = ((j-1)\cdot 2^\ell + 1)h,\\
x_{j,\ell}^e &:= x_{j\cdot 2^\ell} = j \cdot 2^\ell h
\end{aligned}$$

for the start and end nodes of X_j^ℓ. We define $Q_j^\ell := [x_{j,\ell}^s - \frac{h}{2},\ x_{j,\ell}^e + \frac{h}{2}]$, i.e. Q_j^ℓ is the interval that includes X_j^ℓ and is extended by $\frac{h}{2}$ on both ends.

a) Let $\widetilde{\ell} \in \{0, \dots, p\}$ be fixed. Consider for $\eta = 1$ which Q_j^ℓ satisfy the following admissibility conditions ($\mathrm{diam}, \mathrm{dist}$ are defined in the script in (4.10) as follows):

$$\begin{aligned}
&\text{(i)} && \mathrm{diam}_\infty(Q_1^{\widetilde{\ell}}) \le \eta\ \mathrm{dist}(Q_j^\ell, Q_1^{\widetilde{\ell}})\\
&\text{(ii)} && \mathrm{diam}_\infty(Q_j^\ell) \le \eta\ \mathrm{dist}(Q_j^\ell, Q_1^{\widetilde{\ell}})\\
&\text{(iii)} && \min\{\mathrm{diam}_\infty(Q_j^\ell),\ \mathrm{diam}_\infty(Q_1^{\widetilde{\ell}})\} \le \eta\ \mathrm{dist}(Q_j^\ell, Q_1^{\widetilde{\ell}})
\end{aligned}$$

b) Let I_j^ℓ be the node index set corresponding to X_j^ℓ, i.e. $I_j^\ell = \{(j-1)2^\ell + 1, \dots, j2^\ell\}$ with $|I_j^\ell| = 2^\ell$. Then, for all $\ell > 0$ for every j, there exist two unique $j', j'' \in \{1, \dots, 2^{p-\ell+1}\}$ with $I_j^\ell = I_{j'}^{\ell-1} \cup I_{j''}^{\ell-1}$.

We now define the following block partitioning for a matrix $A \in \mathbb{R}^{n\times n}$: Starting from block $I_1^p \times I_1^p$ a block $I_j^\ell \times I_k^\ell$ is subdivided in 4 blocks $I_{j'}^{\ell-1} \times I_{k'}^{\ell-1},\ I_{j''}^{\ell-1} \times I_{k'}^{\ell-1},\ I_{j'}^{\ell-1} \times I_{k''}^{\ell-1},\ I_{j'}^{\ell-1} \times I_{k''}^{\ell-1}$ if the intervals Q_j^ℓ, Q_k^ℓ corresponding to I_j^ℓ, I_k^ℓ do not satisfy the admissibility condition

$$\min\{\mathrm{diam}_\infty(Q_j^\ell),\ \mathrm{diam}_\infty(Q_k^\ell)\} \le \eta\, \mathrm{dist}(Q_j^\ell, Q_k^\ell).$$

(i) Sketch the resulting block partitionings for $p = 4$ and $\eta \in \left\{\frac{1}{2}, 1, 2\right\}$.

(ii) What would be the admissibility condition in order to obtain the matrix format $\mathcal{H}_p$ known from the lecture?

Solution hints:

We have

$$Q_j^\ell = \left[\underbrace{x_{(j-1)\cdot 2^\ell+1}}_{X_{j,\ell}^s} - \frac{h}{2},\ \underbrace{x_{j\cdot 2^\ell}}_{x_{j,\ell}^e} + \frac{h}{2}\right] = \left[\left((j-1)\cdot 2^\ell + \frac{1}{2}\right) h,\ \left(j\cdot 2^\ell + \frac{1}{2}\right) h\right]$$

and

$$\operatorname{diam}_\infty(Q_j^\ell) = \left(j\cdot 2^\ell + \frac{1}{2}\right) h - \left((j-1)\cdot 2^\ell + \frac{1}{2}\right) h = 2^\ell \cdot h.$$

a) We then find

$$Q_1^{\tilde{\ell}} = \left[\frac{h}{2},\ \left(2^{\tilde{\ell}} + \frac{1}{2}\right) h\right].$$

The distance between Q_j^ℓ and $Q_1^{\tilde{\ell}}$ is calculated by the difference between the left boundary node Q_j^ℓ and the right boundary node $Q_1^{\tilde{\ell}}$:

$$\begin{aligned}
\operatorname{dist}(Q_j^\ell, Q_1^{\tilde{\ell}}) &= \min\{\|x - \tilde{x}\|_2 : (x, \tilde{x}) \in Q_j^\ell \times Q_1^{\tilde{\ell}}\} \\
&= \max\left\{x_{j,\ell}^s - \frac{h}{2} - \left(x_{1,\tilde{\ell}}^e + \frac{h}{2}\right),\ 0\right\} \\
&= \max\left\{\left((j-1)\cdot 2^\ell - 2^{\tilde{\ell}}\right) h,\ 0\right\}.
\end{aligned}$$

(i) $\operatorname{diam}_\infty(Q_1^{\tilde{\ell}}) \le \operatorname{dist}(Q_j^\ell, Q_1^{\tilde{\ell}})$ is satisfied for Q_j^ℓ if and only if

$$2^{\tilde{\ell}} \cdot h \le \left((j-1)\cdot 2^\ell - 2^{\tilde{\ell}}\right) h$$

which is equivalent to $j \ge 2^{\tilde{\ell}+1-\ell} + 1$.

(ii) $\operatorname{diam}_\infty(Q_j^\ell) \le \operatorname{dist}(Q_j^\ell, Q_1^{\tilde{\ell}})$ is satisfied for Q_j^ℓ if and only if

$$2^\ell \cdot h \le \left((j-1)\cdot 2^\ell - 2^{\tilde{\ell}}\right) h$$

which is equivalent to $j \ge 2^{\tilde{\ell}-\ell} + 2$.

(iii) $\min\{\operatorname{diam}_\infty(Q_j^\ell), \operatorname{diam}_\infty(Q_1^{\tilde{\ell}})\} \le \operatorname{dist}(Q_j^\ell, Q_1^{\tilde{\ell}})$ is satisfied for Q_j^ℓ if and only if

$$\min\{2^\ell \cdot h, 2^{\tilde{\ell}} \cdot h\} \le \left((j-1)\cdot 2^\ell - 2^{\tilde{\ell}}\right) h$$

which is equivalent to

$$j \ge 2^{\min\{\ell,\tilde{\ell}\}-\ell} + 2^{\tilde{\ell}-\ell} + 1 = \min\{2^{\tilde{\ell}+1-\ell} + 1, 2^{\tilde{\ell}-\ell} + 2\}$$

.

b) We have $\operatorname{diam}_\infty(Q_j^\ell) = \operatorname{diam}_\infty(Q_k^\ell) = 2^\ell \cdot h$ and we obtain $\operatorname{dist}(Q_j^\ell, Q_k^\ell) = 2^\ell h \max\{0, |j - k| - 1\}$ as follows: Let w.l.o.g. $j \leq k$. Then

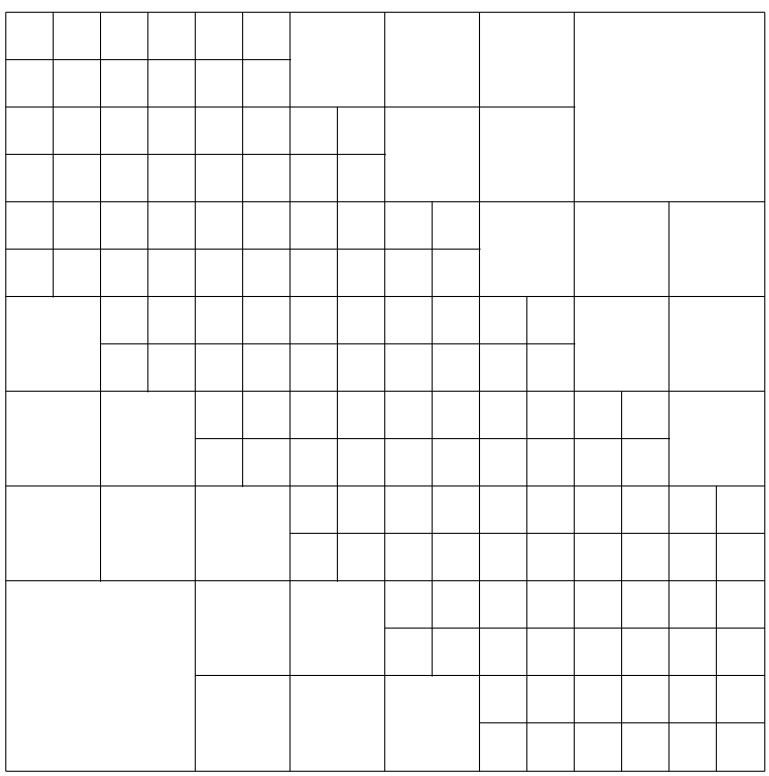
$$\begin{aligned}\operatorname{dist}(Q_j^\ell, Q_k^\ell) &= \max\left\{0, \left((k-1)\cdot 2^\ell + \frac{1}{2}\right)h - \left(j \cdot 2^\ell + \frac{1}{2}\right)h\right\} \\ &= 2^\ell h \max\{0, k-1-j\}.\end{aligned}$$

Hence, $\min\{\operatorname{diam}_\infty(Q_j^\ell),\ \operatorname{diam}_\infty(Q_k^\ell)\} \leq \eta \operatorname{dist}(Q_j^\ell, Q_k^\ell)$ is satisfied for Q_j^ℓ, Q_k^ℓ if and only if

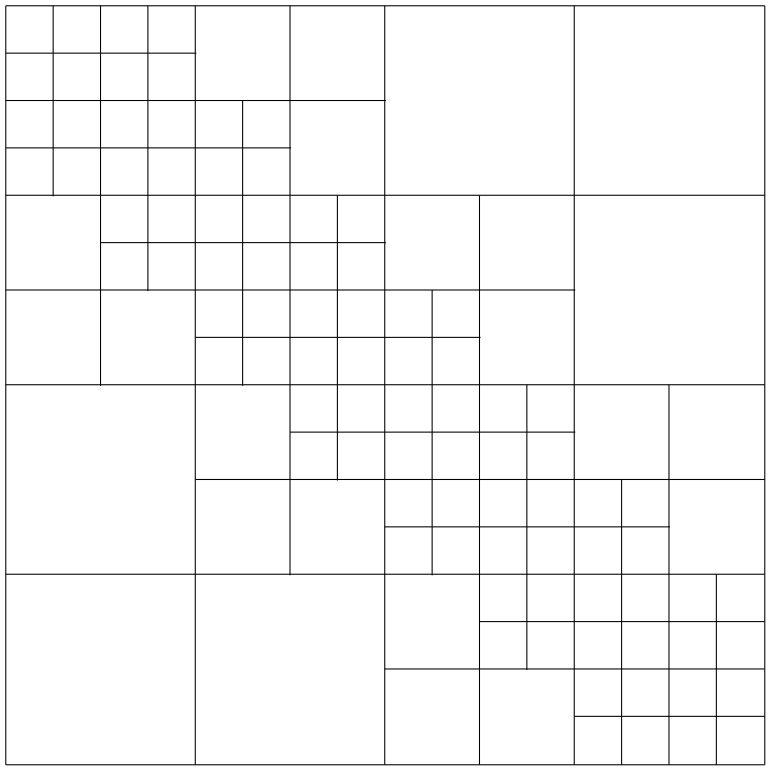
$$\frac{1}{\eta} \leq \max\{0, j-1-k, k-1-j\}$$

.

(i) For $\eta = \frac{1}{2}$, the admissibility condition Q_j^ℓ, Q_k^ℓ is satisfied if and only if $|j-k| \geq 3$. We obtain for $p = 4$:



For $\eta \in \{1, 2\}$ the admissibility condition Q_j^ℓ, Q_k^ℓ is satisfied if and only if $|j-k| \geq 2$. We hence obtain for $p = 4$:



(ii) The admissibility condition $Q_j^\ell \neq Q_k^\ell$ yields the matrix format $\mathcal{H}_p$.

Exercise 2: Let $n \in \mathbb{N}$ and let I be an index set with $\#I = n$. Further, let T be a cluster tree of the index set I with $\#S(t) \neq 1$ for all $t \in T$. Show that

$$\#T \leq 2\ \#L(T) - 1 \leq 2n - 1$$

holds, i.e. $\#T = \mathcal{O}(n)$ holds for the number of vertices of a cluster tree.
Hint: We often abbreviate $v \in T$ to not explicitly write (V, E) and mean that v is a vertex of the tree T. Notations such as $\#T$ are meant analogously.
Solution hints:

Before we prove the actual statement, we briefly look at trees and subtrees. The set of vertices V and the mapping $S\colon V \to \mathcal{P}(V)$ that assigns to each vertex its set of successors identifies a tree $T = (V, E)$ uniquely. Let T' be a tree with corresponding successor mapping S'. Then, T' is a subtree of T if $v \in T$ and $S'(v) \subset S(v)$ for all $v \in T'$. For $v \in T$, let $T(v)$ be the subtree of T given by the vertex set $S^*(v)$, the successors of v, and the mapping $S_{T(v)}$, the restriction of S to $T(v)$ (i.e. v is the root of $T(v)$).

The actual claim is now proven by induction over n. For $n = 1$ follows that $T = L(T) = \{r\}$ and

$$\#T = 1 = 2 \cdot 1 - 1 = 2\ \#L(T) - 1 = 2n - 1,$$

since the label of vertices is always a nonempty index set and $\#S(t) \neq 1$ holds for all $t \in T$. We now assume that the claim also holds for index sets I with $\#I \leq n - 1$ and $n > 1$.
Let T be a cluster tree of a index set I with $\#I = n$ and $n > 1$. W.l.o.g. let $\#S(r) > 1$. Then, T can be split into disjoint subtrees $\{r\}$ and $T(v)$ for all $v \in S(r)$. Hence, $L(T) = \cup_{v \in S(r)} L(T(v))$

is a disjoint union. Since the label of vertices is always a nonempty index set, $\#L(T) \leq n$ holds. With the induction hypothesis, we have $\#T(v) \leq 2\ \#L(T(v)) - 1$ for all $v \in S(r)$. We find

$$\#T = 1 + \sum_{v \in S(r)} \#T(v) \leq 1 + 2\ \#L(T) - \#S(r) \leq 2\ \#L(T) - 1 \leq 2n - 1.$$

Exercises (in class):

Exercise 3: Let $p \in \mathbb{N}$ and $n = 2^p$ points $x_1, \ldots, x_n \in \mathbb{R}$ (i.e. $d = 1$) be given. Futher, let $g\colon \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ be a kernel function. We define the full matrix A by $A := (g(x_i, x_j))_{1 \leq i,j \leq n}$. Complete the method

```
void fill_kernelinterpolation_hp_supermatrix(
      psupermatrix hp, int degree, const double *nodes_x,
      double (*test_function)(int d, const double *x, const double *y));
```

for filling a matrix in $\mathcal{H}_p$ format that approximates the matrix A at the lines marked by */* *Exercise* */* in *supermatrix.c*. The resulting off-diagonal blocks should be constructed by d-variate Chebyshev interpolation of degree `degree` with respect to the first variable of the kernel function. To fill the off-diagonal blocks, you can use the function *interpolate_rkmatrix*. The method

```
psupermatrix new_hp_supermatrix(int p, int k);
```

was extended by a parameter k such that also matrices in $\mathcal{H}_p$ format with ranks $k > 1$ can be constructed for off-diagonal blocks.

Use *example_07.c* for testing and investigate the norms $\|\cdot\|_2$, $\frac{\|\cdot\|_2}{n}$, $\|\cdot\|_\infty$ and the resulting approximation error in the $\frac{n}{2}$th matrix column depending on the number of points n and the degree `degree` of the interpolation. Use the following kernel functions for the tests:

a) $g(x, y) = \exp(-|x - y|^2)$,

b) $g(x, y) = \begin{cases} 0 & : \quad x = y, \\ -\ln|x - y| & : \quad \text{else.} \end{cases}$

Solution hints: